SUPPLEMENTARY MATERIAL OF "WHERE AND WHO?"

Arth J. Shah, Sumasri Talari, Hasini, and Hemant A. Patil

Speech Research Lab, DAU (Formerly DA-IICT)

Algorithm 1 BiLSTM-based Spoof Localization

Require: Frame-level feature sequence $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T]$, frame mask $\mathbf{M}$

Ensure: Frame-level spoof predictions $\hat{\mathbf{y}}$

1: Compute the valid sequence length L from the mask $\mathbf{M}$
2: Pack $\mathbf{X}$ using the valid sequence lengths
3: $\mathbf{H} \leftarrow \mathrm{BiLSTM}(\mathbf{X})$
4: Unpack $\mathbf{H}$ to obtain one hidden representation for each frame
5: $\mathbf{Z} \leftarrow \mathbf{HW} + \mathbf{b}$
6: $\hat{\mathbf{y}}_t \leftarrow \mathrm{Softmax}(\mathbf{Z}_t)$ for $t = 1, \ldots, T$
7: **return** $\hat{\mathbf{y}}$

1. EXPERIMENTAL SETUP

1.1. Localization Model

We use the PartialSpoof dataset, which contains utterances mixing genuine and synthetic speech segments, for our experiments. Each labeled spoof segment's start/end times are converted to frame indices (20 ms per frame at 16 kHz) to create frame-level "spoof" vs "bona fide" labels. We extract feature vectors for each frame using four possible representations: 40-dimensional MFCCs or LFCCs (each augmented with first- and second-order deltas for a total of 120-D), or deep embeddings from self-supervised models (OpenAI's Whisper or HuBERT, producing 512-D features). All features are normalized per utterance to zero mean and unit variance. These frame-by-frame features are fed into a bidirectional LSTM network: this RNN processes the sequence forward and backward and outputs a binary label for each frame (spoof or genuine). We train it with a standard cross-entropy loss on the frame labels, using the Adam optimizer (learning rate = 1e-3, weight decay = 1e-4). Training runs for up to 40 epochs with early stopping (we stop if validation accuracy does not improve for about 8 epochs). In practice, this model learns to assign high probability to frames falling inside the synthetic spoof segment and low probability elsewhere. Algorithm 1 represents the algorithm to regenerate employed classifier.

1.2. Attribution Model

Once a spoofed segment is detected, we classify which attack generated it. Attribution model takes the sequence of frame features for the detected spoof segment and produces a single 256-dimensional "embedding" vector representing that segment. The architecture is based on recently reported Wavelet Integrated Network (WIN) [1]. Concretely, each frame's feature vector is first linearly projected to 64 dimensions and a positional encoding is added. A classification head (using Additive Angular Margin softmax (AAM softmax) as employed in [2]) maps this 256-D embedding to one of the six known attack labels (A01–A06). We set the AAM-Softmax margin to 0.2 and scale to 30. This attribution model is trained similarly (Adam optimizer, lr=1e-3, weight decay=1e-4, batch size=64) for up to 40 epochs with early stopping on validation performance (patience 8 epochs). At testing time, we compare the 256-D embedding of a detected spoof segment with stored reference embeddings (one per attack class) using cosine similarity to perform source verification.

1.3. Motivation for Dataset Regeneration

The proposed framework requires a partially spoofed speech corpus in which each spoofed segment is associated not only with its temporal location and binary spoof/bona fide label, but also with the identity of the generation system responsible for producing that segment. This information is essential for training and evaluating the source attribution stage following spoof localization. Although several partially spoofed speech datasets are publicly available, their released annotations primarily support spoof detection and localization and most of the datasets do not provide the segment-to-generator correspondence required for source attribution.

Among the available resources, the PartialSpoof database is particularly suitable for our experiments, which was specifically designed for partially spoofed speech and provides fine-grained temporal annotations at multiple resolutions. Moreover, the employed version (of PartialSpoof dataset) in spoof diarization study [3], employs only 6 known attacks, i.e., A01-A06 from ASVSpoof 2019 dataset as reported in [4]. However, as per authors best knowledge and belief, there exist no such sub-track of PartialSpoof dataset [5], specifically carefully designed for open-set partial spoof attribution task.

We therefore regenerated the partially spoofed data by following the construction procedure of the original PartialSpoof database [5], while introducing only one additional

bookkeeping step: the identity of the unknown generation system in testing sub-track associated with each selected spoof segment was retained throughout the construction process. All other major aspects of the original construction procedure, including normalization, VAD-based segment selection, substitution, temporal annotation, and spoof-ratio-based balancing, were retained as closely as possible. Thus, the regenerated corpus preserves the characteristics of the original PartialSpoof data while providing the additional segment-level source information required by the proposed source-attribution task.

1.4. Review of Existing Partially Spoofed Datasets

We reviewed existing partially spoofed speech datasets with respect to their suitability for segment-level open-set source attribution. The primary requirement was that a dataset should provide both fine-grained temporal information indicating whether a region is spoofed and the identity of the generation system responsible for each spoofed region. The reviewed datasets include HAD (Half-Truth) [**?**], ADD 2022 Partially Fake [**?**], ADD 2023 Partially Fake, and PartialSpoof [5].

HAD replaces isolated words, including named entities such as persons, locations, organizations, and time expressions, with TTS-generated speech. However, the released annotations provide the fake or real status of the segments without specifying the TTS or voice-conversion system responsible for each replacement [**?**]. Similarly, the ADD 2022 Partially Fake track replaces partial regions of genuine utterances with real or synthesized speech, but the generation algorithms associated with the spoofed regions are not provided in a form suitable for segment-level source attribution [**?**]. ADD 2023 extends the partially fake setting with manipulated-region localization and additional noise and format-conversion conditions; however, the generator identity of each spoofed region is still not available as a segment-level annotation. PartialSpoof dataset, which serves as a baseline and is the closest match to the requirements of this work. It constructs partially spoofed utterances by substituting segments between bona fide and also provides annotations for close set partial-spoof attribution task. The underlying ASVspoof 2019 LA attack identities are available for the source utterances and are used in attack-specific analyses, however, only A01-A06 attacks are employed in eval set as mentioned in [4]. Therefore, the limitation of the reviewed resources is the absence of partially spoofed speech or temporal annotations or segment attribution, but the absence of the explicit unknown segment-to-generator correspondence required by the real-life based source-attribution task as considered objective of this work. This forced us to regenerate the PartialSpoof data while preserving the generation identity of every substituted spoof segment.

1.5. Dataset Construction

We followed the five-stage construction procedure described in the PartialSpoof database [5], applied to the ASVspoof 2019 LA corpus [6]. The construction procedure was retained with minimal modification. The only additional information maintained throughout the process was the identity of the generation system associated with each spoof donor segment. This modification allows the regenerated corpus to retain the original PartialSpoof formulation while providing segment-level source labels for the proposed source-verification experiments.

1.5.1. Step 1: Normalization and Voice Activity Detection

Bona fide and spoofed utterances from the ASVspoof 2019 LA corpus were first amplitude-normalized to -26 dBov following the normalization procedure used in PartialSpoof. This normalization reduces amplitude differences between the source and substituted speech that could otherwise provide unintended cues for detecting the manipulated regions. Voice activity detection (VAD) was subsequently applied to identify speech-active regions that could be considered for segment substitution. The resulting VAD regions were used as candidate speech segments for the subsequent selection process.

1.5.2. Step 2: Segment Selection

For each candidate region, a segment from the opposite class was selected while preserving the speaker identity and maintaining a comparable duration. Specifically, bona fide segments were substituted with spoofed segments and spoofed segments were substituted with bona fide segments, where applicable, using speech from a different utterance of the same speaker. Three constraints were maintained during selection: (i) the donor segment was selected from a different utterance of the same speaker; (ii) the same donor segment was not reused more than once within a target utterance; and (iii) the duration of the source and substituted regions was matched.

For spoofed donor segments, the corresponding ASVspoof 2019 LA attack identity was retained as metadata at this stage. Hence, in addition to the temporal location and binary class of each substituted region, the generation-system identity was explicitly associated with the spoofed segment. This information was carried forward throughout the remaining dataset construction process.

1.5.3. Step 3: Substitution and Concatenation

The selected segments were substituted into the target utterances using the same concatenation procedure as the original PartialSpoof construction. Time-domain cross-correlation was used to determine suitable concatenation points, followed by overlap-add processing to reduce discontinuities at

the boundaries between the original and substituted speech. This procedure preserves the partially spoofed characteristics of the original database while minimizing audible artifacts introduced by the segment replacement process [5].

1.5.4. Step 4: Temporal and Source Labeling

Each regenerated utterance was annotated at the same fine-grained temporal resolutions used in PartialSpoof, ranging from 20 ms to 640 ms. In addition to an utterance-level label, each temporal unit was assigned a binary label indicating bona fide or spoof content. A temporal unit containing generated content was labeled as spoof, while the remaining regions were labeled as bona fide.

In addition to these binary labels, the attack identity associated with each spoofed donor segment was retained. Consequently, each spoofed region contains both its temporal boundaries and its corresponding generation-system identity. The resulting annotation can therefore be represented as

$$\mathcal{L}_k = \left(t_k^{(s)}, t_k^{(e)}, y_k, g_k \right), \tag{1}$$

where $t_k^{(s)}$ and $t_k^{(e)}$ denote the start and end times of the k-th region, $y_k \in \{\text{bona fide}, \text{spoof}\}$ denotes its binary class, and g_k denotes the generation-system identity for a spoofed region.

1.5.5. Step 5: Balancing and Post-processing

Following the original PartialSpoof construction, the regenerated utterances were grouped according to their intra-speech generated-segment ratio, defined as the proportion of speech duration occupied by spoofed segments. The utterances were quantized into ten spoof-ratio levels and randomly sampled to provide a balanced representation of low- and high-spoof-ratio conditions. This balancing procedure was retained to minimize changes in the distribution of partially spoofed utterances relative to the original PartialSpoof construction [5].

Note: The total number of samples for train/dev/test subtrack of dataset remains identical to ASVSpoof 2019, and Partial spoof dataset [3, 6].

1.6. Attack Identity and Evaluation Data

The ASVspoof 2019 LA corpus contains multiple spoofing attack identities. In the PartialSpoof evaluation setting, the spoofing methods considered for attack-specific analysis correspond to attacks A07–A19 [5]. The original spoof diarization study groups only known attacks (A01-A06) according to their availability during training, development, and evaluation set (as reported in [4]).

Use of Generative AI Disclosure

Generative AI tools were used for language polishing and bug fixes. All the outputs were manually reviewed and verified by the authors.

2. REFERENCES

[1] Arth J. Shah, Aniket Pandey, and Hemant A. Patil, "Winning against audio deepfakes," *IEEE Signal Processing Letters*, vol. 33, pp. 2240–2244, 2026.

[2] V Negroni, D Salvi, P Bestagini, S Tubaro, et al., "Source verification for speech deepfakes," in *INTERSPEECH*, pp. 1548–1552. 2025, Rotterdam, Netherlands.

[3] Lin Zhang, Xin Wang, Erica Cooper, Mireia Diez, Federico Landini, Nicholas Evans, and Junichi Yamagishi, "Spoof diarization:" what spoofed when" in partially spoofed audio," *INTERSPEECH*, pp. 502–506, 2024, Kos Island, Greece.

[4] Kyo-Won Koo, Chan-yeong Lim, Jee-Weon Jun, Hye-Jin Shim, and Ha-Jin Yu, "Token-based attractors and cross-attention in spoof diarization," in *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, 2025, Honolulu, Hawaii, USA, pp. 1–7.

[5] Lin Zhang, Xin Wang, Erica Cooper, Nicholas Evans, and Junichi Yamagishi, "The PartialSpoof database and countermeasures for the detection of short fake speech segments embedded in an utterance," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 813–825, 2022.

[6] Andreas Nautsch et al., "ASVSpoof 2019: Spoofing countermeasures for the detection of synthesized, converted, and replayed speech," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 3, no. 2, pp. 252–265, 2021.